

## BLAIR, NE, USA

WMO: 720405

Lat: 41.412N

Lon: 96.109W

Elev: 396

StdP: 96.66

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00436

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -19.0      | -16.7 | -23.8                         | 0.5 | -18.2 | -21.3 | 0.6 | -15.6 | 13.4                  | -5.1 | 12.3 | -4.4 | 3.4                   | 320  | 0.554 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.8                    | 33.9            | 23.8 | 32.4 | 23.5 | 31.0 | 23.1 | 25.9                | 31.1 | 25.0 | 30.1 | 24.0 | 29.3 | 5.4                  | 170  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 24.4                            | 20.3 | 29.7 | 23.2 | 18.8 | 28.4 | 22.4 | 18.0 | 27.7 | 82.9          | 31.2 | 78.0 | 30.2 | 74.5 | 29.2 | 29.1              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |      |            |      |    |    |
|-------------------|------|-----|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|------|------------|------|----|----|
|                   |      |     | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |      | n=50 years |      |    |    |
| 1%                | 2.5% | 5%  | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max  | Min        | Max  | DB | WB |
| (a)               | (b)  | (c) | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |    |    |
| (4) 11.8          | 10.6 | 9.1 | -23.6                      | 36.5 | 2.7                | 1.6 | -25.6                                              | 37.6 | -27.1      | 38.6 | -28.6      | 39.5 | -30.5      | 40.7 |    |    |
| (5)               |      |     | -24.0                      | 27.5 | 2.6                | 0.8 | -25.8                                              | 28.1 | -27.3      | 28.5 | -28.8      | 29.0 | -30.7      | 29.5 |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 11.0   | -4.3  | -2.9  | 5.1   | 10.9  | 17.2  | 23.0  | 24.5  | 23.1  | 19.7  | 12.0  | 4.7   | -2.3  |  |  |
|                                                                        | DBStd       | 11.57  | 6.69  | 6.69  | 6.83  | 5.59  | 5.20  | 3.42  | 3.20  | 2.90  | 4.28  | 5.17  | 5.97  | 6.32  |  |  |
|                                                                        | HDD10.0     | 1647   | 444   | 363   | 181   | 57    | 6     | 0     | 0     | 0     | 1     | 38    | 176   | 381   |  |  |
|                                                                        | HDD18.3     | 3316   | 702   | 595   | 411   | 229   | 85    | 4     | 1     | 2     | 34    | 203   | 409   | 639   |  |  |
|                                                                        | CDD10.0     | 2001   | 0     | 2     | 31    | 83    | 229   | 389   | 449   | 407   | 292   | 101   | 17    | 1     |  |  |
|                                                                        | CDD18.3     | 628    | 0     | 0     | 2     | 6     | 50    | 143   | 192   | 151   | 76    | 8     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 5672   | 0     | 0     | 27    | 77    | 471   | 1339  | 1851  | 1219  | 617   | 69    | 3     | 0     |  |  |
|                                                                        | CDH26.7     | 2030   | 0     | 0     | 5     | 20    | 165   | 503   | 721   | 412   | 191   | 13    | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 4.3    | 4.5   | 4.6   | 5.1   | 5.4   | 4.7   | 4.1   | 3.2   | 3.2   | 3.7   | 4.2   | 4.7   | 4.4   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 770    | 16    | 15    | 50    | 75    | 104   | 101   | 96    | 84    | 91    | 63    | 34    | 23    |  |  |
|                                                                        | PrecMax     | 1060   | 41    | 62    | 134   | 181   | 229   | 224   | 245   | 201   | 358   | 136   | 130   | 113   |  |  |
|                                                                        | PrecMin     | 542    | 0     | 0     | 0     | 9     | 14    | 0     | 0     | 16    | 0     | 0     | 0     | 0     |  |  |
|                                                                        | PrecStd     | 151    | 12    | 13    | 42    | 45    | 49    | 59    | 57    | 57    | 67    | 41    | 29    | 22    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 13.2  | 18.4  | 26.3  | 29.4  | 34.1  | 35.5  | 36.2  | 35.0  | 33.2  | 27.8  | 22.7  | 14.6  |  |  |
|                                                                        |             | MCWB   | 6.4   | 11.2  | 15.2  | 17.2  | 21.5  | 23.3  | 24.3  | 22.5  | 23.0  | 17.4  | 13.6  | 9.7   |  |  |
|                                                                        | 2%          | DB     | 9.5   | 12.7  | 22.5  | 25.3  | 30.8  | 32.9  | 33.8  | 32.5  | 31.2  | 25.1  | 19.0  | 11.3  |  |  |
|                                                                        |             | MCWB   | 5.5   | 7.3   | 13.2  | 14.9  | 19.7  | 23.3  | 24.8  | 23.8  | 22.6  | 16.0  | 12.3  | 7.7   |  |  |
|                                                                        | 5%          | DB     | 6.8   | 9.4   | 18.9  | 22.6  | 28.0  | 31.4  | 32.3  | 30.7  | 28.9  | 22.7  | 16.2  | 8.8   |  |  |
|                                                                        |             | MCWB   | 3.5   | 5.8   | 12.2  | 13.7  | 18.3  | 22.7  | 24.8  | 23.5  | 21.7  | 14.7  | 10.6  | 5.4   |  |  |
|                                                                        | 10%         | DB     | 4.4   | 6.2   | 15.9  | 20.0  | 26.0  | 29.7  | 30.5  | 28.9  | 27.0  | 20.5  | 13.2  | 6.1   |  |  |
|                                                                        |             | MCWB   | 2.1   | 3.5   | 10.3  | 12.7  | 17.4  | 21.8  | 24.1  | 22.4  | 20.2  | 14.0  | 8.4   | 3.5   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB     | 7.4   | 12.0  | 17.4  | 18.5  | 22.6  | 26.4  | 27.5  | 26.8  | 24.7  | 20.1  | 15.3  | 11.7  |  |  |
|                                                                        |             | MCDB   | 12.2  | 18.7  | 22.3  | 25.4  | 28.7  | 32.1  | 32.9  | 32.0  | 30.6  | 24.1  | 19.6  | 13.0  |  |  |
|                                                                        | 2%          | WB     | 5.4   | 8.3   | 15.3  | 16.7  | 21.3  | 25.0  | 26.3  | 25.2  | 23.6  | 18.2  | 12.9  | 7.8   |  |  |
|                                                                        |             | MCDB   | 8.7   | 12.3  | 20.3  | 23.1  | 28.0  | 30.7  | 31.6  | 29.8  | 29.0  | 23.1  | 18.3  | 10.0  |  |  |
|                                                                        | 5%          | WB     | 3.9   | 5.6   | 12.7  | 15.3  | 20.1  | 23.9  | 25.4  | 24.2  | 22.8  | 16.6  | 11.1  | 5.5   |  |  |
|                                                                        |             | MCDB   | 6.5   | 9.1   | 18.2  | 21.0  | 26.3  | 30.0  | 30.5  | 29.0  | 27.5  | 20.6  | 15.5  | 8.6   |  |  |
|                                                                        | 10%         | WB     | 2.3   | 3.7   | 10.2  | 13.8  | 18.8  | 22.9  | 24.4  | 23.2  | 21.8  | 14.8  | 8.9   | 3.7   |  |  |
|                                                                        |             | MCDB   | 4.5   | 6.4   | 16.4  | 18.9  | 24.4  | 28.3  | 29.5  | 27.8  | 25.6  | 19.4  | 13.3  | 6.2   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR   | 9.2   | 9.2   | 10.7  | 11.2  | 10.8  | 10.2  | 9.8   | 9.9   | 10.9  | 11.0  | 10.1  | 8.4   |  |  |
|                                                                        |             | MCDBR  | 12.5  | 13.3  | 16.4  | 15.4  | 13.9  | 11.8  | 10.7  | 11.7  | 12.6  | 14.2  | 13.7  | 12.4  |  |  |
|                                                                        | 5% WB       | MCWBR  | 8.8   | 8.7   | 8.8   | 7.9   | 5.6   | 5.4   | 5.3   | 5.4   | 5.7   | 7.2   | 7.9   | 8.7   |  |  |
|                                                                        |             | MCWBR  | 11.1  | 12.2  | 13.8  | 12.6  | 11.9  | 10.4  | 9.6   | 10.1  | 10.4  | 11.7  | 12.6  | 11.4  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.280 | 0.313 | 0.329 | 0.387 | 0.411 | 0.415 | 0.425 | 0.415 | 0.375 | 0.330 | 0.296 | 0.283 |  |  |
|                                                                        | taud        |        | 2.445 | 2.376 | 2.396 | 2.246 | 2.251 | 2.298 | 2.266 | 2.298 | 2.368 | 2.484 | 2.504 | 2.455 |  |  |
|                                                                        | Ebn at Noon |        | 881   | 896   | 921   | 882   | 867   | 861   | 848   | 846   | 859   | 868   | 857   | 846   |  |  |
|                                                                        | Edh at Noon |        | 76    | 96    | 106   | 132   | 135   | 130   | 132   | 125   | 108   | 86    | 72    | 70    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 1.99  | 2.94  | 3.80  | 4.88  | 5.79  | 6.44  | 6.50  | 5.59  | 4.61  | 3.22  | 2.24  | 1.67  |  |  |
|                                                                        | RadStd      |        | 0.19  | 0.17  | 0.34  | 0.35  | 0.43  | 0.50  | 0.42  | 0.38  | 0.30  | 0.38  | 0.15  | 0.17  |  |  |

## Historical Trends

|                              |       | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            | DBAvg | (d)     | (e)    | (f)     | (g)   | (h)   | (i)         | (j)     | (k)     | (l)     |
| (47) Regional (39 neighbors) |       | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |

Nomenclature: See separate page